

Counterdiabatic Hamiltonian Monte Carlo

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Abstract

Counterdiabatic Hamiltonian Monte Carlo (CHMC) combines the principles of counterdiabatic driving from quantum control theory with Sequential Monte Carlo (SMC) methods to enable efficient sampling from complex probability distributions. By learning a gauge potential that compensates for the non-equilibrium effects of time-varying Hamiltonians, CHMC achieves faster convergence and better exploration compared to standard HMC methods. This poster presents the theoretical foundations and practical implementation of CHMC.

Hamiltonian Dynamics

Hamiltonian dynamics describe the evolution of a physical system in terms of position $q \in \mathbb{R}^d$ and momentum $p \in \mathbb{R}^d$ coordinates. The dynamics are governed by a *Hamiltonian* function $H(q, p)$ through Hamilton's equations:

$$\dot{q} = \frac{\partial H}{\partial p} \quad (1)$$

$$\dot{p} = -\frac{\partial H}{\partial q} \quad (2)$$

The Hamiltonian typically decomposes as $H(q, p) = T(p) + V(q)$, where $T(p) = \frac{1}{2}|p|^2$ is the kinetic energy and $V(q)$ is the potential energy. The flow ϕ_H^t generated by these equations preserves the Hamiltonian: $H(\phi_H^t(q, p)) = H(q, p)$ for all t .

[Figure placeholder: Phase space trajectories showing Hamiltonian flow]

Statistical Mechanics & Equilibrium

In statistical mechanics, a system in thermal equilibrium at inverse temperature β has a probability distribution over phase space given by the *canonical ensemble*:

$$\rho_H(q, p) = \frac{1}{Z} e^{-\beta H(q, p)} \quad (3)$$

where $Z = \int e^{-\beta H(q, p)} dq dp$ is the partition function. For $\beta = 1$, this becomes $\rho_H(q, p) \propto e^{-H(q, p)}$.

The key property is that Hamiltonian dynamics preserve this equilibrium distribution: if $(q, p) \sim \rho_H$, then $\phi_H^t(q, p) \sim \rho_H$ for all t . This makes Hamiltonian dynamics ideal for MCMC sampling, as the marginal distribution on q is $\pi(q) \propto \int e^{-H(q, p)} dp = e^{-V(q)}$.

[Figure placeholder: Equilibrium distribution visualization]

Counterdiabatic Gauge Potential

When the Hamiltonian depends on time through a parameter $\lambda(t)$, we have $H(\lambda(t))$. The naive approach of following the time-dependent Hamiltonian flow $\phi_{H(\lambda(t))}$ fails because:

$$\partial_t \rho_{H(\lambda(t))} = \{H, \rho_{H(\lambda(t))}\} + \dot{\lambda} \partial_\lambda \rho_{H(\lambda(t))} \neq 0 \quad (4)$$

The distribution *lags behind* the changing Hamiltonian. The *counterdiabatic gauge potential* $A(q, p)$ solves this by satisfying:

$$\{A, H\} = \partial_\lambda H - \langle \partial_\lambda H \rangle \quad (5)$$

where $\{A, H\} = \frac{\partial A}{\partial q} \frac{\partial H}{\partial p} - \frac{\partial A}{\partial p} \frac{\partial H}{\partial q}$ is the Poisson bracket. Using the modified Hamiltonian $H_c = H + \dot{\lambda} A$, the flow $\phi_{H_c}^t$ perfectly preserves the time-varying equilibrium: $\phi_{H_c}^t \rho_{H(\lambda(t))} = \rho_{H(\lambda(t+\tau))}$.

[Figure placeholder: Comparison of naive vs counterdiabatic evolution]

Sequential Monte Carlo

Sequential Monte Carlo (SMC) methods sample from a target distribution π_1 by constructing a sequence of intermediate distributions π_λ for $\lambda \in [0, 1]$, where π_0 is easy to sample from and π_1 is the target.

At each step k , we:

1. Evolve particles using a Markov kernel K_{λ_k} that targets π_{λ_k}
2. Compute importance weights: $w_i^{(k)} \propto w_i^{(k-1)} \frac{\pi_{\lambda_{k+1}}(x_i)}{\pi_{\lambda_k}(x_i)}$
3. Resample when effective sample size (ESS) drops below a threshold

The key challenge is designing kernels K_λ that efficiently transport particles between distributions. Standard HMC kernels require many steps to equilibrate, leading to slow mixing.

[Figure placeholder: SMC particle evolution and resampling]

SMC with Learned Transport Map

Instead of using generic MCMC kernels, we can learn a *transport map* that directly transforms particles from π_{λ_k} to $\pi_{\lambda_{k+1}}$. This is more efficient but requires:

- A parameterized family of maps $T_\phi(x; \lambda)$
- A loss function to learn ϕ that measures how well T_ϕ transports the distribution
- Samples from the current distribution to train on

Counterdiabatic driving provides a natural transport map: the flow $\phi_{H_c}^t$ generated by $H_c = H + \dot{\lambda} A$ exactly transports $\rho_{H(\lambda(t))}$ to $\rho_{H(\lambda(t+\Delta t))}$. The gauge potential A acts as the learned transport map, parameterized by an ansatz $A_\phi(q, p)$.

[Figure placeholder: Learned transport map visualization]

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Learning the Gauge Potential: At each SMC step with parameter λ_k , we learn A_ϕ by minimizing:

$$\mathcal{L}(A_\phi) = \mathbb{E}_{(q, p) \sim \rho_{H(\lambda_k)}} \left[(\{A_\phi, H\} - \partial_\lambda H + \langle \partial_\lambda H \rangle)^2 \right] \quad (6)$$

This is done using gradient descent on samples from the current distribution, with A_ϕ parameterized as a polynomial or neural network.

The CHMC Algorithm:

1. Initialize particles from π_0
2. For each λ_k in schedule:
 - Fit A_ϕ using current particles
 - Evolve particles using $H_c = H(\lambda_k) + \dot{\lambda}_k A_\phi$ via leapfrog integrator
 - Compute importance weights
 - Resample if ESS drops
3. Return final particles from π_1

Key Advantages:

- Faster convergence: counterdiabatic driving eliminates lag
- Better exploration: learned transport adapts to distribution geometry
- Preserves Hamiltonian structure: maintains volume preservation

[Figure placeholder: CHMC algorithm performance comparison]

References